

PRASHANTH KUMAR KADASI

Data Analyst & AI Systems Engineer

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PROFESSIONAL SUMMARY

Data Analyst & AI Systems Engineer with 3+ years of experience delivering high-throughput analytics, discrete allocation optimization, and production autonomous multi-agent AI ecosystems. Expert in Python, SQL, BigQuery, and modern BI alongside LLM fine-tuning (20B models on AMD MI300X), Model Context Protocol (FastMCP 2024-11-05 spec), edge computer vision pipelines, and RAG architectures. Successfully engineered an 18,000+ candidate MBBS discrete allocation engine validated against official KNRUHS 2026 Phase 1 results (within 2 preferences and 73 ranks of cutoff), an autonomous 10-agent multi-agent swarm with daily self-evolving target benchmarks, and 18 sector intelligence dashboards for US & UK enterprise clients.

KEY QUANTIFIED ACHIEVEMENTS

- 18K+ candidate MBBS discrete allocation simulator validated against KNRUHS Phase 1 (within 2 choices)
- 20B parameter LLM fine-tuned on AMD MI300X (+51% recommendation steerability improvement)
- 60% reduction in manual analytics reporting time through end-to-end BigQuery pipeline automation
- 10-Agent autonomous AI swarm with FastMCP tools and daily self-evaluating target fitness benchmarks
- 18 sector intelligence dashboards deployed with automated weekly AI insights for enterprise clients
- 15+ dashboards & 30+ analytical reports delivered for US & UK stakeholders across 3 years

PROFESSIONAL EXPERIENCE

Black Piano Mar 2026 – Present

Data Analyst | Remote

- Deployed 18 sector intelligence dashboards along with end-to-end automated data pipelines using Google AppSheet, BigQuery, and Looker Studio for 4 enterprise clients.
- Built and integrated weekly AI insight generation that automatically produces sector-specific market analysis for each of the 18 dashboards.
- Migrating all data pipelines into Google Cloud Platform (GCP) with scheduled BigQuery queries for enterprise-grade scalability, auditability, and cost efficiency.
- Continuing analytics delivery for Pi Datametrics client post-transition, enhancing enterprise reporting and real-time visualization systems.

Pi Software Solutions Pvt Ltd (Pi - Datametrics) Mar 2023 – Feb 2026

Data Analyst | Remote

- Delivered 15+ production dashboards and 30+ analytical reports across elections, brands, and market performance for US & UK enterprise clients.
- Built automated data pipelines using BigQuery, AppScript, and Python — reducing manual reporting time by 60% and enabling automated daily data refreshes.
- Developed Pi-API Python package for automated BigQuery uploads/downloads, deployed as a web service on Render for cross-functional analytics workflows.
- Conducted NLP sentiment analysis on large-scale election datasets and built predictive time-series models (ARIMA, LSTM) for market trend forecasting.
- Automated dashboard generation across 20+ client accounts spanning Insurance, Gambling, and E-commerce sectors.

Independent AI Systems Research & Development Jan 2024 – Present

AI Systems Engineer | Remote

- Architected AI Eco, an autonomous 10-agent AI swarm operating daily via GitHub Actions with living memory compaction (<4,000 words), FastMCP JSON-RPC/SSE protocol interfaces, and self-evaluating target fitness benchmarks.
- Engineered Retail Shelf Intelligence (Solari Platform), integrating an edge contour-segmentation vision engine with cloud VLM fallback for planogram compliance, out-of-stock detection, and discrete product allocation.
- Fine-tuned GPT-OSS-20B on AMD MI300X GPUs to evaluate brand recommendation steerability, boosting recommendation rates from 25.5% to 76.5% (+51% improvement) under rigorous A/B evaluation (Hugging Face Hub).
- Engineered mSeat, an MBBS discrete allocation simulator for 18,000+ candidates; validated against official KNRUHS 2026 Phase 1 results predicting actual allotment within 2 choices (and 73 ranks of cutoff) via O(1) multi-quota indexing and 90.6% dataset compression (545 KB).
- Developed MyLocalCLI, a local-first agentic coding assistant supporting 6 AI providers (Gemini, Claude, OpenAI, Ollama, NVIDIA NIM, OpenRouter), 26 tools, and 5 sub-agents with zero remote data leakage.

TECHNICAL SKILLS

AI & Machine Learning:	LLM Fine-Tuning (LoRA/QLoRA on AMD MI300X), Autonomous Multi-Agent Swarms, Model Context Protocol (FastMCP 2024-11-05), Edge Computer Vision, Discrete Optimization, RAG Architectures, Gemini / Claude / OpenAI APIs
Data & SQL Engineering:	SQL (Expert), Google BigQuery, Python (Pandas, NumPy, PyTorch), Data Modeling, Automated ETL Pipelines, Database Indexing
BI & Visualization:	Looker Studio, Tableau, Power BI, Plotly, Chart.js, Real-Time Interactive Dashboards, Trend Forecasting
Cloud, Platforms & DevOps:	Google Cloud Platform (GCP), Vercel Serverless, Docker, GitHub Actions CI/CD, Git, Linux/Bash, ROCm

KEY ENGINEERING PROJECTS

AI Eco — Autonomous 10-Agent Swarm & FastMCP Server — Python, Multi-Agent Swarms, FastMCP, GitHub Actions, Living Memory

Autonomous 10-agent multi-agent swarm operating daily scheduled pipelines to ingest commits, compute velocity telemetry, evaluate fitness benchmarks, and expose real-time portfolio tools over standard Model Context Protocol (FastMCP JSON-RPC 2.0 & SSE transports). Features bounded living memory (<4,000 words), automated debt ledger pruning, adversarial resilience checks, and daily dual-audience technical logs.

🔗 Live Dashboard: kprsnt.in/ecosystem • 📄 Swarm Logs: kprsnt.in/ecosystem/logs • 📄 GitHub: github.com/kprsnt2/kprsnt.in

Retail Shelf Intelligence — Solari Autonomous Platform — Edge Computer Vision, Heuristic Contours, Cloud VLM, Discrete Optimization, Python

Real-time retail inventory intelligence system featuring a dual-mode vision engine (edge CPU contour segmentation + cloud VLM fallback) for planogram compliance, out-of-stock detection, and discrete product allocation under real-world store lighting and camera angles. Designed for cost-efficient local execution with cloud escalation.

🔗 Portfolio: kprsnt.in/projects • 📄 GitHub: github.com/kprsnt2/retail_shelf_intelligence

mSeat — MBBS Discrete Allocation Simulator & MCP — JavaScript, Discrete Optimization, FastMCP Server, Combinatorial Algorithms, Vercel

High-performance discrete allocation engine simulating 18,000+ candidates across 59 medical colleges under a complex 7D reservation matrix (Local/Unreserved, Gender, EWS, Social Categories). Validated against official KNRUHS 2026 Phase 1 results, matching the allotted college within a 2-preference delta and 73 ranks of cutoff. Employs O(1) multi-quota cumulative indexing and 90.6% dataset compression (545 KB).

🔗 Live Application: mseat.kprsnt.in • 📄 GitHub: github.com/kprsnt2/mSeat

BrandXY — LLM Recommendation Steerability Research (20B Model) — GPT-OSS-20B, Hugging Face, AMD MI300X, PyTorch, LoRA

Fine-tuned 20B parameter open-source LLM on AMD MI300X (192GB HBM3) to evaluate brand recommendation steerability. Boosted target recommendation rates from 25.49% baseline to 76.47% (+51% improvement) under rigorous multi-round A/B evaluation across 100+ simulated consumer personas. Published model weights on Hugging Face Hub; arXiv paper in progress.

🗨️ HuggingFace: [brandXY-chat](https://huggingface.co/brandXY-chat) • 📄 GitHub: github.com/kprsnt2/brand-llm-finetune-oss-20b

MyLocalCLI — Local-First Agentic Coding Assistant — Node.js, CLI, 6 AI Providers, 26 Specialized Tools, 5 Sub-Agents

Privacy-first terminal coding assistant orchestrating Gemini, Claude, OpenAI, Ollama, NVIDIA NIM, and OpenRouter across 26 specialized tools, 5 autonomous agents, and 22 skill modules. Features full bash execution, AST analysis, and zero-telemetry local-first guarantees.

🔗 Live Portal: mlc.kprsnt.in • 📄 GitHub: github.com/kprsnt2/MyLocalCLI

18 Sector Intelligence Dashboards & Automated Pipelines — Google BigQuery, Looker Studio, Google Cloud Platform (GCP), AppScript, AI Insights

Enterprise-scale analytics platform delivering automated dashboards with weekly AI-generated market insights across 18 sectors for 4 enterprise clients. Automated data pipelines with App Script and BigQuery cut manual reporting time by 60%. Currently migrating all pipelines into GCP scheduled workflows.

🔗 Live Dashboard: dashboard.kprsnt.in • 📄 GitHub: github.com/kprsnt2/dashboard_site

Drug Discovery GPT-20B — Pharmaceutical Molecular Interpretation — GPT-OSS-20B, PyTorch, AMD MI300X, FDA Orange Book, openFDA, PubChem

Fine-tuned 20B parameter LLM on pharmaceutical datasets (FDA Orange Book — 40K+ drugs, openFDA, PubChem) for molecular SMILES structure interpretation, chemical property extraction, ADMET property prediction, and drug repurposing research.

🗨️ HuggingFace: [drug-discovery-gpt-20b](https://huggingface.co/drug-discovery-gpt-20b) • 📄 GitHub: github.com/kprsnt2/drug_discovery

EDUCATION

Anurag Group of Institutions

M. Pharmacy — Pharmaceutical Analysis & Quality Assurance
Jawaharlal Nehru Technological University Hyderabad (JNTUH) | May 2012

PROFILES & PORTALS

Portfolio:	kprsnt.in	LinkedIn:	linkedin.com/in/prashanth-kumar-kadasi
GitHub:	github.com/kprsnt2	HuggingFace:	huggingface.co/kprsnt